Demian Kong

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GitHub | Personal Website
PhD Applications • LLM Agents • Research Systems

Research Profile

Bioinformatics M.S. graduate and AI systems practitioner whose research and engineering work spans scientific data systems, natural-language data access, Text2SQL, tool use, and configurable agent frameworks. Currently focused on reliable LLM agents and general-purpose AI systems, with Chrona as an application for exploring planning, scheduling, controlled execution, review, and recovery.

Research Interests

- Reliable LLM agents, tool learning, and multi-step execution systems
- Representation learning from language to structured actions, adaptive policies, and world models
- Agent–environment feedback, planning, and evaluation; scientific automation and software-controlled applications

Education

University of Chinese Academy of Sciences (UCAS), M.S. in Bioinformatics, 2023–2026, Beijing
Graduated June 2026. M.S. thesis: Large Language Model–Based Retrieval of Bioinformatics Data and Development of a Configurable Agent Framework.
Jilin University, B.S. in Pharmacy, 2019–2023, Jilin

Research Experience

National Genomics Data Center (NGDC), China National Center for Bioinformation (CNCB), Master' s Researcher, 2023–June 2026, Beijing | Advisor: Wenming Zhao

- Built conversational retrieval agents for genomic and epigenomic databases with LangGraph and FastAPI; designed multi-stage Text2SQL and interactive data-exploration workflows for EWAS AI Assistant.
- Contributed to the multi-omics pipeline and chatbot for iDog, and curated and integrated biomarker data from more than 20 public resources for BioKA.

Jilin University, Undergraduate Thesis Research, 2022–2023

- Established a CCl_4-induced liver-fibrosis model and investigated MKP5-related mechanisms using retroviral overexpression, RNA-seq, GO/KEGG enrichment, qRT-PCR, Western blotting, and histological analysis.

Jilin University, Undergraduate Research, 2021–2022

- Conducted organic-compound synthesis and crystallographic research; determined absolute configuration using single-crystal X-ray diffraction.

Selected Projects

Chrona — Local-first AI work executor | Creator | Mar 2026–Present | github.com/saya-ashen/Chrona

- Turns tasks into planned, scheduled, executable, and inspectable work through a Task → Plan → Schedule → Execute → Review/Recover loop.
- Bun/TypeScript monorepo integrating React, Hono, SQLite/Prisma, a graph runtime, model-provider adapters, MCP integrations, and a CLI; v0.2.0, currently in alpha/local development.

Dingent — Open-source framework for general-purpose LLM agents | Lead Architect and Maintainer through June 2026 | GitHub

- Built a LangGraph backend, Next.js conversational interface, and React administration interface supporting modular plugins, visual configuration, and development workflows.
- Transferred the project to the team upon graduation; currently provide technical support only.

EWAS AI Assistant — Natural-language data-access system for epigenomics | Primary Developer through June 2026 | NGDC BioAgent

- Built a LangGraph workflow spanning intent parsing, SQL generation, data retrieval, and LLM-based synthesis, with interactive result visualization through FastAPI and Vue 3.
- M.S. thesis result: entity-enhanced retrieval and a generate–verify–correct loop achieved 76.7% Text2SQL execution accuracy on complex queries.

Selected Publications

1. Yang F, Xiong Z, Zong W, **Kong D**, et al. EWAS Open Platform 2026: a deeply integrated resource for epigenome-wide association studies. Nucleic Acids Research. 2026;54(D1):D1061–D1068. doi:10.1093/nar/gkaf1155.
2. Liu Y †, Wang Y †, Sun J †, **Kong D †**, et al. iDog: a multi-omics resource for canids study. Nucleic Acids Research. 2025;53(D1):D1039–D1046. doi:10.1093/nar/gkae1031.
3. Wang Y, Lin Y, Wu S, Sun J, Meng Y, **Kong D**, et al. BioKA: a curated and integrated biomarker knowledgebase for animals. Nucleic Acids Research. 2024;52(D1):D1121–D1130. doi:10.1093/nar/gkad873.
4. CNCB–NGDC Members and Partners. Database Resources of the National Genomics Data Center, China National Center for Bioinformation in 2025. Nucleic Acids Research. 2025;53(D1):D30–D44. doi:10.1093/nar/gkae978. (**Kong D** listed among consortium members.)
5. CNCB–NGDC Members and Partners. Database Resources of the National Genomics Data Center, China National Center for Bioinformation in 2026. Nucleic Acids Research. 2026;54(D1):D28–D47. doi:10.1093/nar/gkaf1172. (**Kong D** listed among consortium members.)

† Equal contribution.

Technical Skills

- **Programming & Data:** Python, JavaScript/TypeScript, SQL, Bash, Pandas, NumPy
- **AI & Agents:** PyTorch, Hugging Face Transformers, LangChain/LangGraph, MCP, DeepEval
- **Applications & Systems:** FastAPI, React, Next.js, Vue.js, Docker, Linux, Git
- **Languages:** Chinese (native); English (IELTS 7.0)

Honors

- Provincial Undergraduate Innovation and Entrepreneurship Training Program, Jilin University.